

In [4]: `import pandas as pd`

In [5]: `dataset=pd.read_csv('insurance_pre.csv')`

In [6]: `dataset`

Out[6]:

	age	sex	bmi	children	smoker	charges
0	19	female	27.900	0	yes	16884.92400
1	18	male	33.770	1	no	1725.55230
2	28	male	33.000	3	no	4449.46200
3	33	male	22.705	0	no	21984.47061
4	32	male	28.880	0	no	3866.85520
...	...	...	...	...	...	...
1333	50	male	30.970	3	no	10600.54830
1334	18	female	31.920	0	no	2205.98080
1335	18	female	36.850	0	no	1629.83350
1336	21	female	25.800	0	no	2007.94500
1337	61	female	29.070	0	yes	29141.36030

1338 rows × 6 columns

In [7]: `dataset=pd.get_dummies(dataset,drop_first=True)`

In [8]: `dataset`

Out[8]:

	age	bmi	children	charges	sex_male	smoker_yes
0	19	27.900	0	16884.92400	0	1
1	18	33.770	1	1725.55230	1	0
2	28	33.000	3	4449.46200	1	0
3	33	22.705	0	21984.47061	1	0
4	32	28.880	0	3866.85520	1	0
...	...	...	...	...	...	...
1333	50	30.970	3	10600.54830	1	0
1334	18	31.920	0	2205.98080	0	0
1335	18	36.850	0	1629.83350	0	0
1336	21	25.800	0	2007.94500	0	0
1337	61	29.070	0	29141.36030	0	1

1338 rows × 6 columns

```
In [9]: dataset.columns
```

```
Out[9]: Index(['age', 'bmi', 'children', 'charges', 'sex_male', 'smoker_yes'], dtype='object')
```

```
In [12]: independent=dataset[['age', 'bmi', 'children', 'sex_male', 'smoker_yes' ]]
```

```
In [13]: independent
```

```
Out[13]:
```

	age	bmi	children	sex_male	smoker_yes
0	19	27.900	0	0	1
1	18	33.770	1	1	0
2	28	33.000	3	1	0
3	33	22.705	0	1	0
4	32	28.880	0	1	0
...	...	...	...	...	...
1333	50	30.970	3	1	0
1334	18	31.920	0	0	0
1335	18	36.850	0	0	0
1336	21	25.800	0	0	0
1337	61	29.070	0	0	1

1338 rows × 5 columns

```
In [14]: dependent=dataset[['charges']]
```

```
In [15]: dependent
```

```
Out[15]: 0      16884.92400
1       1725.55230
2       4449.46200
3      21984.47061
4       3866.85520
...
1333    10600.54830
1334     2205.98080
1335     1629.83350
1336     2007.94500
1337    29141.36030
Name: charges, Length: 1338, dtype: float64
```

```
In [19]: from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(independent,dependent,test_size=0.30,random_state=0)
```

```
In [20]: from sklearn.linear_model import LinearRegression
regressor=LinearRegression()
regressor.fit(x_train,y_train)
```

```
Out[20]: LinearRegression(copy_X=True, fit_intercept=True, n_jobs=None, normalize=False)
```

```
In [21]: weight=regressor.coef_
weight
```

```
Out[21]: array([ 257.8006705 ,  321.06004271,  469.58113407,  -41.74825718,
                23418.6671912 ])
```

```
In [22]: intercept=regressor.intercept_
intercept
```

```
Out[22]: -12057.244845998523
```

```
In [23]: y_pred=regressor.predict(x_test)
```

```
In [24]: from sklearn.metrics import r2_score
r_score=r2_score(y_test,y_pred)
```

```
In [25]: r_score
```

```
Out[25]: 0.7894790349867009
```

```
In [27]: import pickle
filename='finalized_model_Mul_linear.sav'
```

```
In [28]: pickle.dump(regressor,open(filename,'wb'))
```

```
In [29]: loaded_model=pickle.load(open('finalized_model_Mul_linear.sav','rb'))
result=loaded_model.predict([[35,21.77,2,1,1]])
```

```
In [30]: result
```

```
Out[30]: array([28271.33695325])
```

```
In [ ]:
```